Gold in Biological Environments: A Martini Model

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Abstract

Gold is used in a variety of biomedical and drug delivery applications, yet it is difficult to reliably model gold interactions with biological materials in the Martini force field. In a previous study[1], we showed that aggregation of existing Martini nanoparticle models used to represent gold nanoparticles depends on the choice of gold-core parameters. However, a Martini gold bead model that simultaneously accounts for gold–gold and gold–biomolecule surface affinities has not been developed. Here we develop a Martini coarse-grained gold bead designed as a counterpart to the INTERFACE force field. The model was designed to preserve the structural, energetic, and interfacial properties of gold. To parameterize the gold model, we match several targets to their atomistic or experimental counterparts, including gold-gold self-interaction energies, gold-biomolecule adsorption free energies. We then performed unbiased nanoparticle dimer simulations initialized without aggregation. These simulations produced a dimer with interparticle spacings qualitatively consistent with those observed in experimental cryo-EM images. We will further test the model through quantitative comparisons of simulated and experimental interparticle-distance distributions. We also plan to formalize the parameterization framework for extension to other metals.

Background

- Gold (GNPs, Figure 1) is used in biosensing, biolabelling, and controlled drug delivery
- In these applications, gold interacts with biological molecules.
- The INTERFACE force field[2] is an additive inorganic force field that can accurately capture metal-biomolecule interactions.
- The ability to capture metal-biological molecule interactions in Martini[3] is currently not possible
- Researchers have used adhoc polar bead[5], hydrophobic bead[6], and soft sphere[7] models that do not fully capture gold behavior

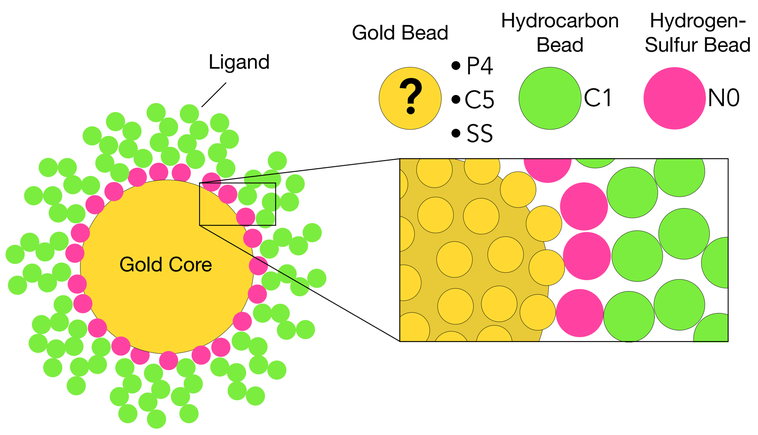


Fig 1. Representation of gold nanoparticle model core and assigned bead types

Model Goals

Develop a Martini model for gold that accurately captures gold-gold and gold biomolecule interactions

1. What is the best mapping between a gold cluster and a Martini Bead?
2. Can we reproduce the dimerization free energies of atomistic gold clusters?
3. Can we reproduce the interaction free energies between gold surfaces and small molecules in aqueous environments?

Approach

- Match size of gold clusters to a Martini bead
- Overlay Martini bead with an adjustable virtual site
- Match gold-gold dimerization free energies between atomistic system and coarse-grained
- Match adsorption free energies for amino acids to a gold slab and compare them between atomistic and coarse-grained systems
- Match adsorption free energies for lipids to a gold slab and compare them between atomistic and coarse-grained systems

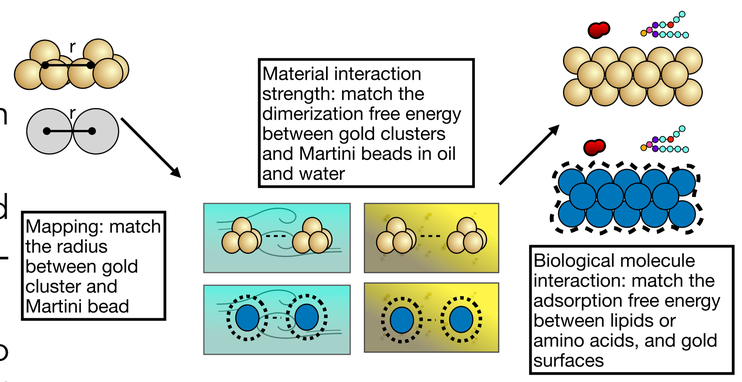


Fig 2. Martini coarse-grained gold model road map

Results

Determining gold cluster size and dimerization free energy in solvents

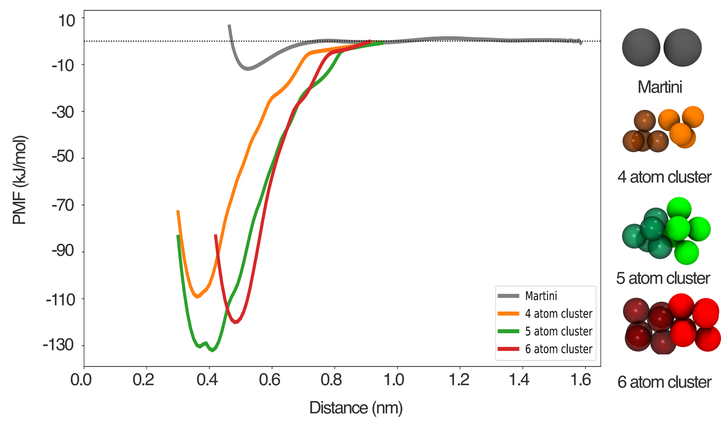


Fig 3. **Dimerization free energy profiles** between Martini S beads (grey), or atomistic gold clusters of varying sizes: 4-atom cluster (orange), 5-atom cluster (green), 6-atom cluster (red). Representative images of clusters are colored as above.

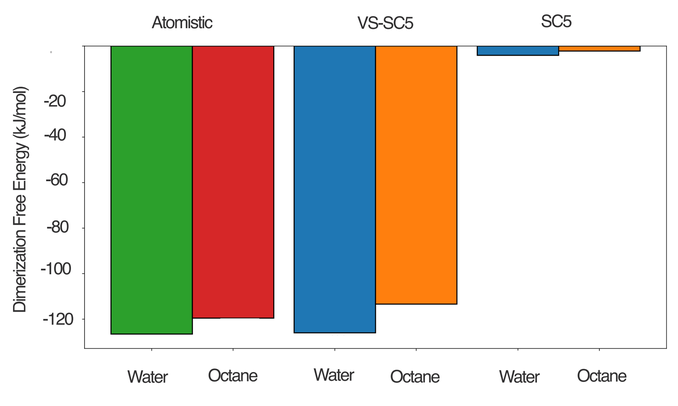


Fig 4. **Dimerization free energy** between gold clusters (6 atom), Martini beads overlayed with virtual sites (vs-SC5), or non-virtual site Martini beads (SC5). Dimerization free energies were calculated under water solvent or octane solvent conditions.

Small molecule adsorption free energies used to tune the gold model's behavior in biological environments

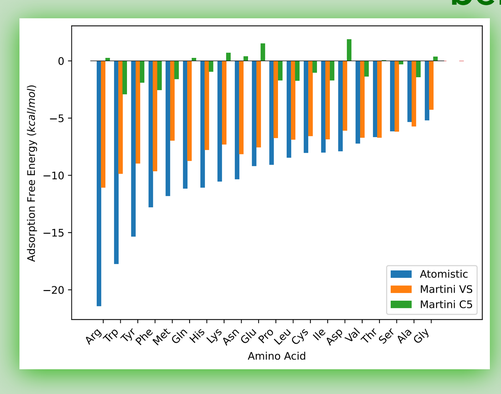


Fig 5. **Adsorption free energy of the 20 amino acids to a slab** of: atomistic gold (blue), Martini bead overlayed with virtual site tuned to newly specified potentials for Martini gold (orange), or Martini C5 beads (green). Molecular image from a simulation of an amino acid (red) and a gold slab (gold) in a water box (cyan).

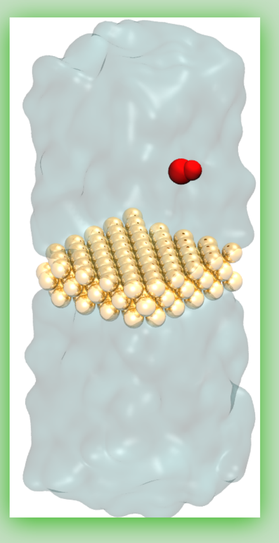
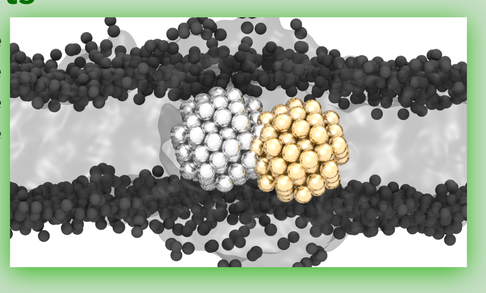


Fig 6. **Representative image** of gold nanoparticle model built with the adjusted Martini virtual site gold potentials.



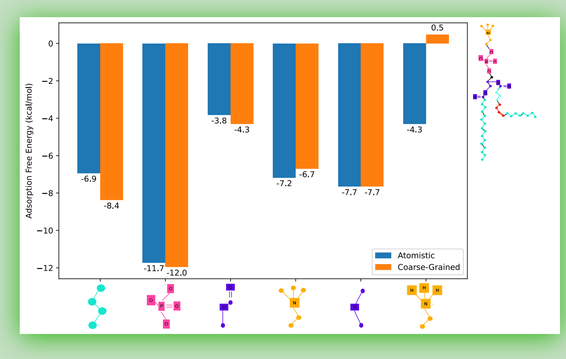


Fig 7. **Adsorption free energy of lipid pieces to a slab** of: atomistic gold (blue), Martini bead overlayed with virtual site tuned to newly specified potentials for Martini gold (orange)

Approach

- Simulation software: Gromacs 2024[4]
- Force Field: CHARMM[5] with Interface[2], Martini[3,8]
- System Building Tools: Charmm-gui[9,11], COBY[13], Insane[10]
- Free energy profiles were obtained with COLVARS[14]
- All atom systems were equilibrated for 20 ns and production was 200 ns
- All coarse-grained systems were equilibrated for 200 ns and production was 2 μs

Summary

- Here we have developed the first Martini gold bead that captures gold-gold and gold-biomolecule interactions.
- We match several targets to appropriately model gold cluster size, dimerization propensity, and small molecule adsorption.
- We tested the model by building a gold nanoparticle system and observed strong gold nanoparticle dimerization, which is expected from qualitative observation (Fig. 6, 8).

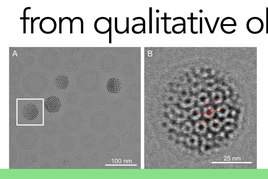


Fig 8. Cryo-TEM images of GNPs aggregating in polymersomes[15]

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Check out our paper showing that the nanoparticle core parameters affect aggregation!!